Version 2 Upgrade Project Specification – Board -P25

1. Background

The current P25 board is designed for three-phase electrical measurement using an AFE (Analog Front-End) circuit based on the ADE9000. The existing design is functional but has several limitations in terms of power supply redundancy, compatibility with different wiring configurations (Wye and Delta), and electrical robustness.

2. Problem Statement

The existing design relies on a single-phase + neutral input for powering the system. This introduces a vulnerability in case of phase loss or neutral disconnection. Additionally, the hardware is optimized for Wye (with neutral) wiring but lacks full compatibility with Delta (without neutral) systems. Electrical robustness and EMC/EMI protections are also limited.

3. Objectives

- Implement a redundant power supply capable of operating even if one or two phases are lost and functional without a neutral connection.
- Increase electrical robustness and safety.
- Support both Wye (star) and Delta wiring configurations with minimal hardware changes.
- Maintain compatibility with the ADE9000 measurement chain.

4. Proposed Solution & Architecture

The proposed design will integrate a RAC05-K/480 high-voltage AC/DC power module fed through a three-phase diode bridge with adequate overvoltage, inrush, and EMI protections. The architecture will be designed to ensure phase redundancy and support for both Wye and Delta systems, with proper grounding separation between AGND (analog) and DGND (digital/SPI).

Voltage Input (L1, L2, L3, N)

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Protection (Fuse + MOV)

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Rectifier bridges 8 diodes (L1, L2, L3, N)

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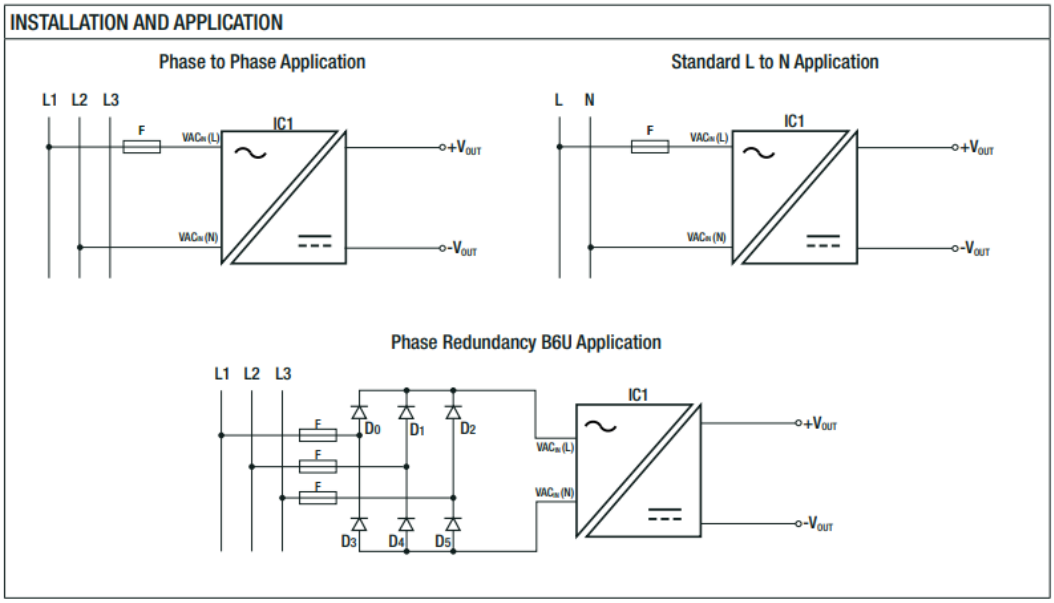
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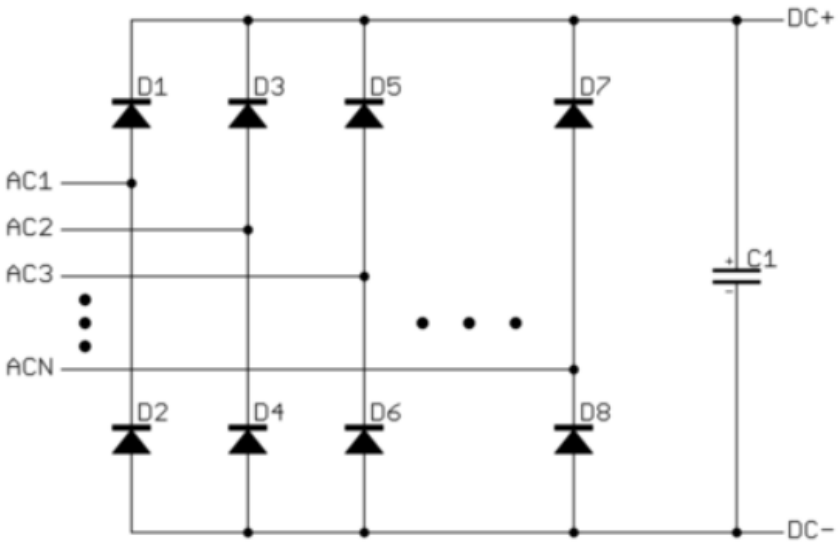
RAC05-K/480 (DC Input)

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Voltage Output 5V → MCU + GSM + AFE



Here is a generalized full wave rectifier for N AC inputs:



5. Comparative Design Options

Option	Advantages	Drawbacks

Single-phase power supply	Simple, low cost	Loss of power on phase failure
Three-phase redundant power supply	Phase loss tolerance, compatible with Wye & Delta	Higher BOM cost, more complex design

6. Critical Components

- RAC05-K/480 AC/DC power module
- Three-phase bridge rectifier
- MOV surge protection
- Common-mode choke
- Ferrite beads for AGND/DGND separation
- TVS diodes on measurement inputs

7. Input Mode Compatibility

The new design must handle:
- L1-N, L2-N, L3-N (Wye)
- L1-L2, L2-L3, L3-L1 (Delta)
- Mixed configurations

AFE scaling and filtering must remain accurate under both wiring modes.

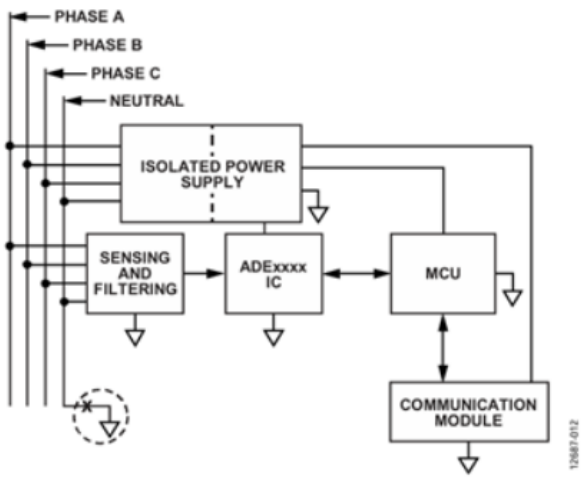


Figure 2. 3P4W metering system isolation requirement.

8. Functional Requirements

- Power output: 5V DC / 5W
- Operation with 1, 2, or 3 phases connected
- Measurement accuracy per ADE9000 datasheet

- Isolation and safety compliance with IEC standards
- EMC/EMI compliance per EN 55032/EN 61000

9. Conclusion & Expected Benefits

This upgrade will significantly improve the reliability and flexibility of the Sensing3Ph product, making it compatible with a wider range of electrical systems and more resilient to power disturbances.